An Implementation of Simplicial Dynamic Epistemic Logic in Python

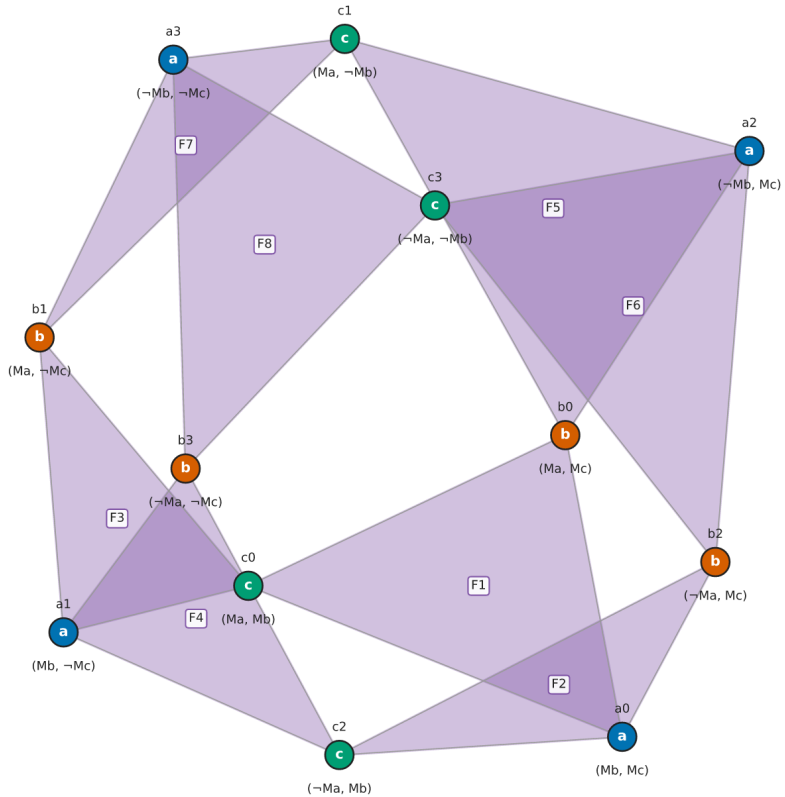
Natalia Mejía Bautista, Universidad Nacional Autónoma de México
Philip Sink

October 20, 2026

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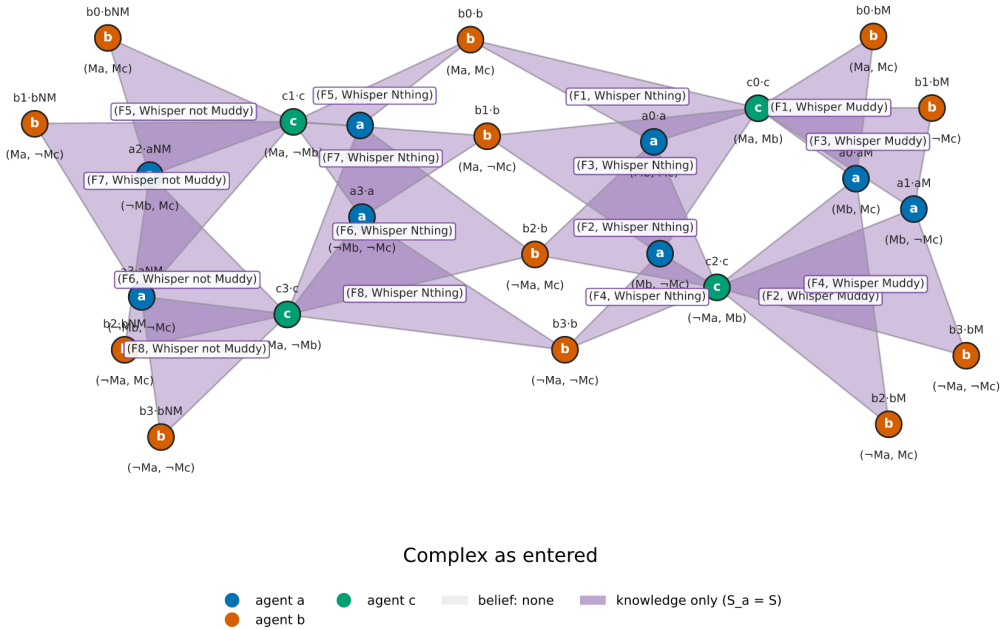
1. Introduction
2. D.E.L. Simplicially
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Outline



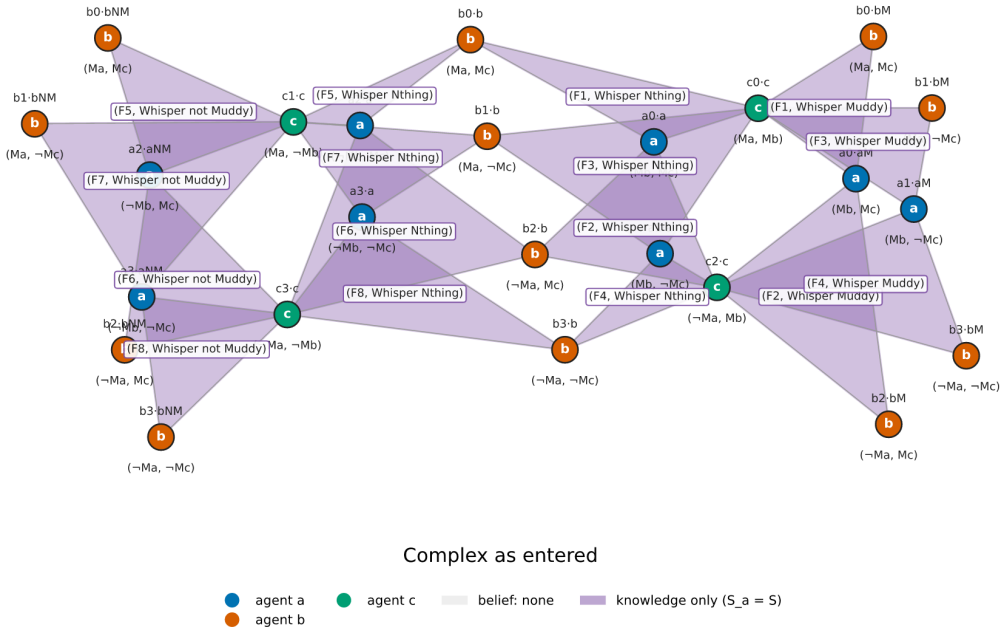
The simple explanation for what this project is/has done is implement the definitions from my thesis in an environment like Python

Why Simulations?



I believe there are two audiences for simulations of D.E.L. as we are undertaking

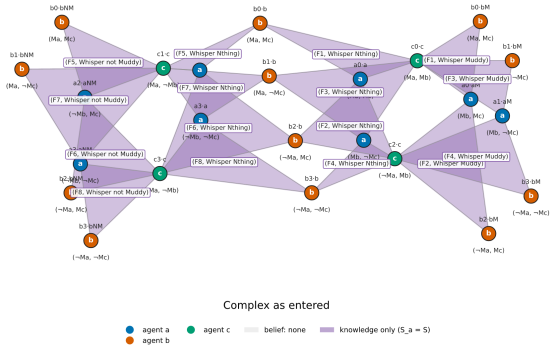
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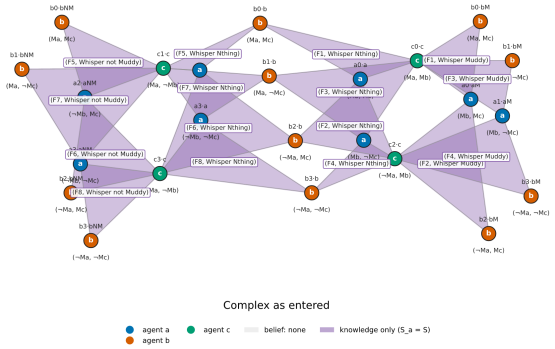
The first is computer scientists: Insofar as D.E.L. is a language for describing distributed protocols[GLR21; Dit+22; Att+14; PS18; VCR21; Cas+23], there is reason to simulate D.E.L. in order to simulate said protocols.

Why Philosophers?



There are epistemic questions I think a framework like this might be able to tackle in a novel way.

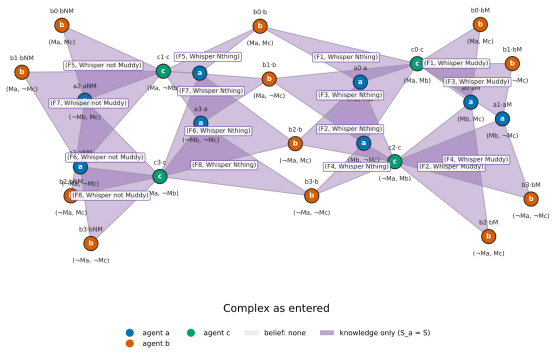
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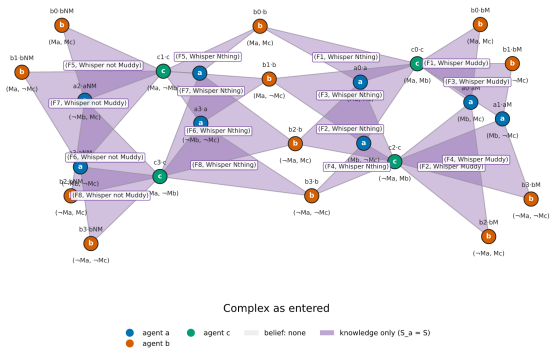


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Addressing this question would require a tool where one can compare the same "scenario" with revision switched on in one case and off in another.

My ask: Can this setup address this meaningfully? What pitfalls should I expect? What other questions might this framework address (not necessarily involving revision)?

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An important point of contrast is a possible world, which is typically modeled as a complete set of literals.

Example: Muddy Children Part 1



Imagine you are the parent of three children, a, b, and c. You are a good parent, so these children are perfect reasoners.

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After playing outside $m > 0$-many of the children are muddy.

You sit the children in a circle. They can therefore observe the muddy status of every other child, but of course not themselves.

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Your perspective consists of the muddy status of b and c.

Hence, simple combinatorial reasoning tells us there are four possible perspectives for each child.

Uncertainty in Simplicial Semantics

We now need to model the fact that each child is uncertain about whether or not they themselves are muddy.

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The idea behind uncertainty is then very simple. Suppose two sets of perspectives have a nonempty intersection.

Agents whose perspectives are in this intersection *have the same perspective* in these sets. Hence, they cannot tell them apart, and are uncertain between these two sets.

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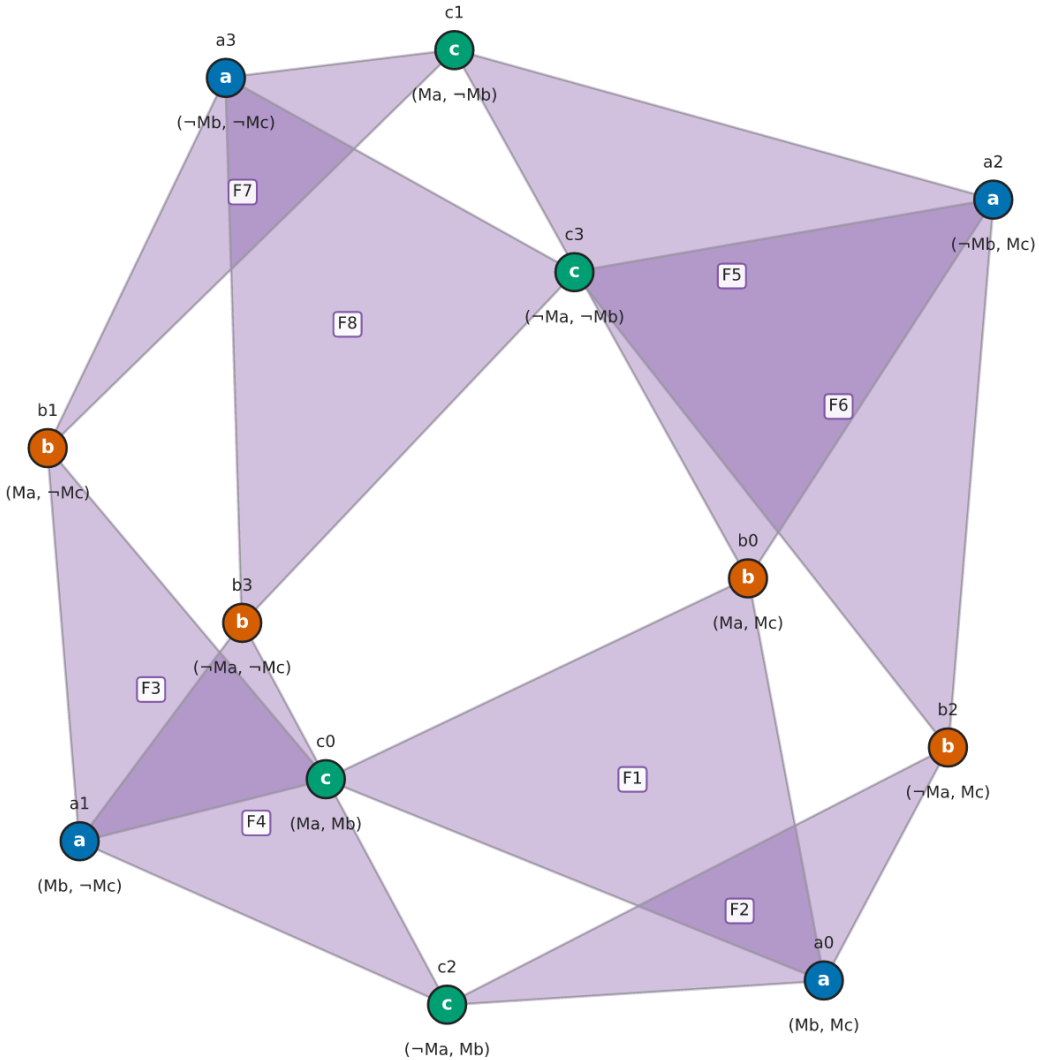
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Placing yourself in a's perspective, the question is, which of b and c's perspectives do you consider possible simultaneously?

The answer (for this example) is fortunately simple: They're seeing information consistent with yours, plus they have to *agree* as to whether or not you are muddy.

This corresponds, in this setup, to all consistent *uniquely colored* sets, e.g., those with exactly one perspective per agent.

Example: Muddy Children Part 2



Simplicial complex

agent a agent b agent c belief: none knowledge only (S_a = S)

Dynamics

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Intuitively, this announcement correspond to each child learning $M_a \vee M_b \vee M_c$, learning every other child learns this fact, and so on.

We represent that with "update perspectives" as follows, and call this an *action*.

Public Announcement Update

In particular, this action is a "public announcement": there is one perspective per agent, and every agent considers that perspective correct in the update.

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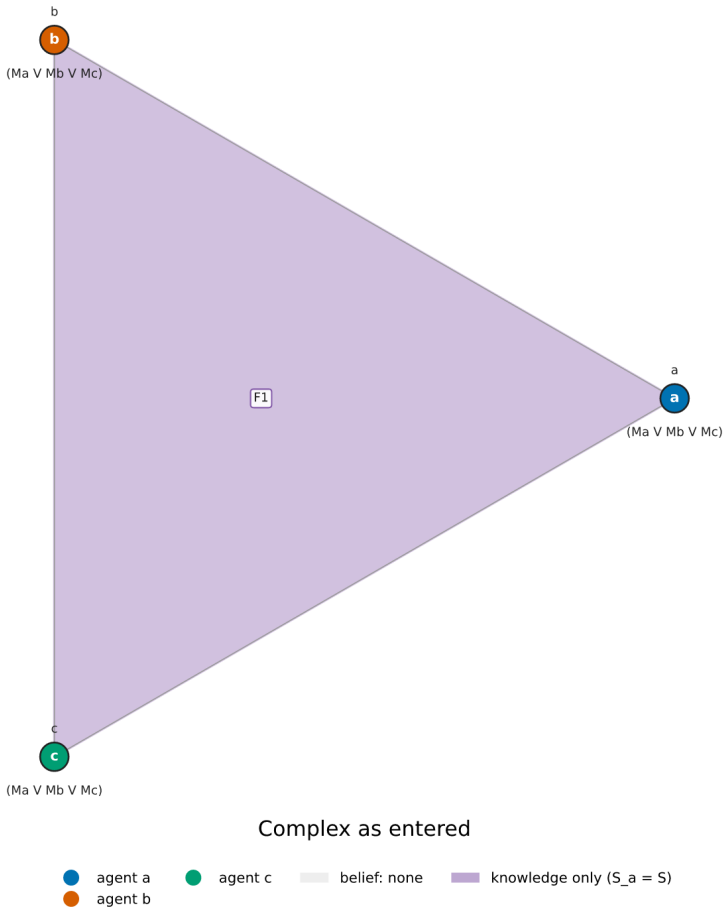
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The idea with actions is that they can be combined with inputs to give outputs using a restriction of the Cartesian product.

What's true at a perspective in an action is what **has** to be not false[2] in an input perspective for this action to apply/change it. This is called a *precondition*.

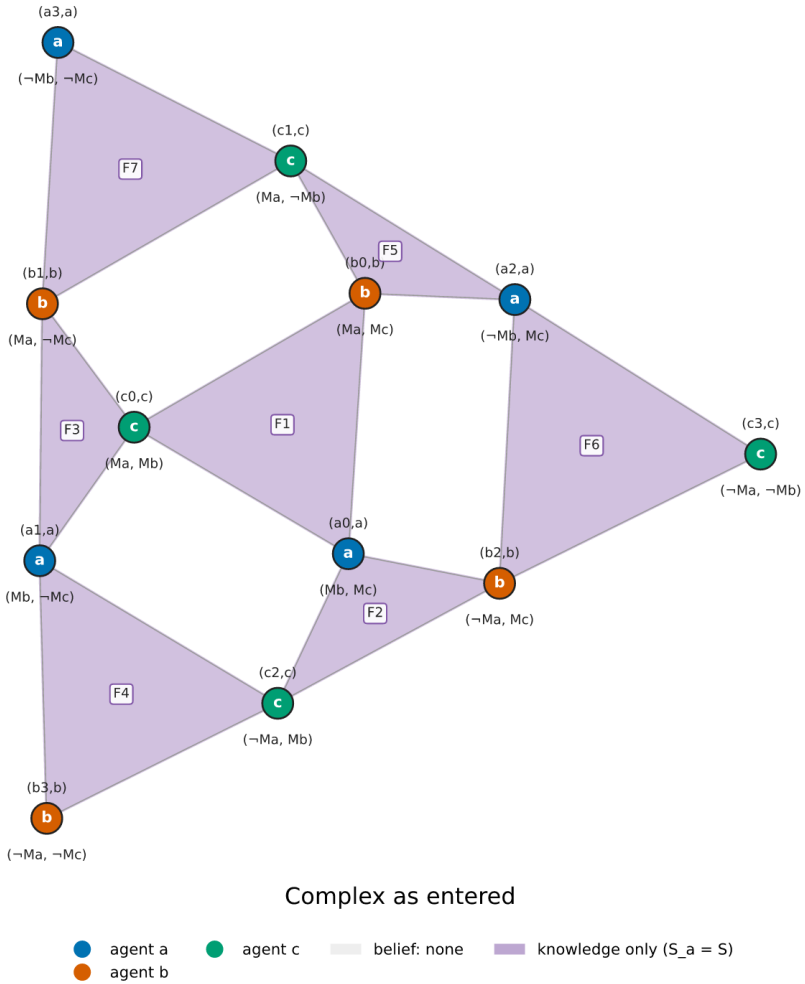
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Public Announcement Update



Public Announcement Update

If we match preconditions to the input, and draw facets when the first and second projections appear in both the input and action respectively, we get the following output.



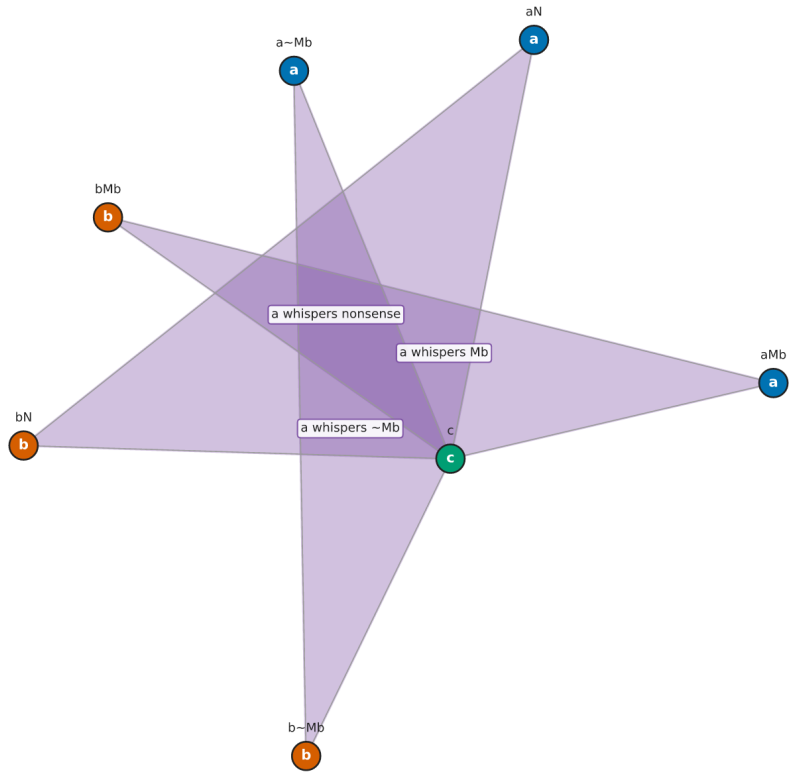
Subtler actions

Suppose a whispers something to b. c observes this, and knows that a is not a liar.

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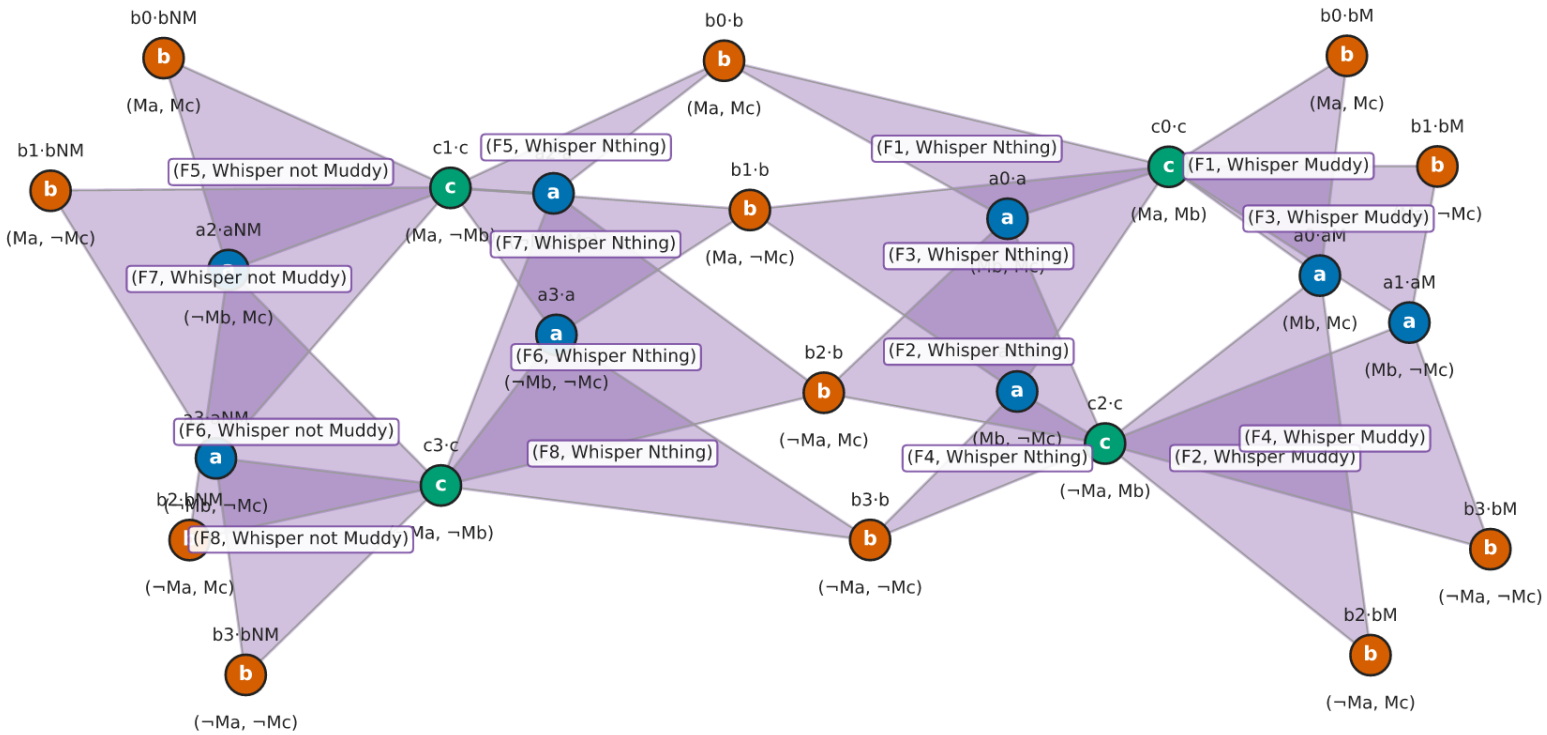
Suppose *a* whispers something to *b*. *c* observes this, and knows that *a* is not a liar.

What does this action look like?



Subtler Update

Now let's apply this action to the same input.



False Beliefs

So far, the children are perfect reasoners and have no misconceptions. What if that were not the case?

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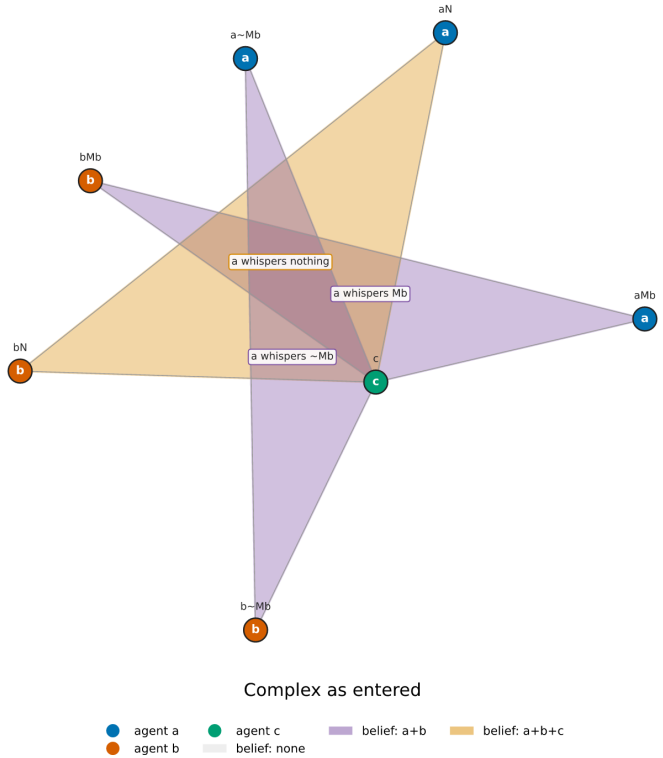
So, agents are less uncertain, but potentially incorrect!

Actions with False Beliefs

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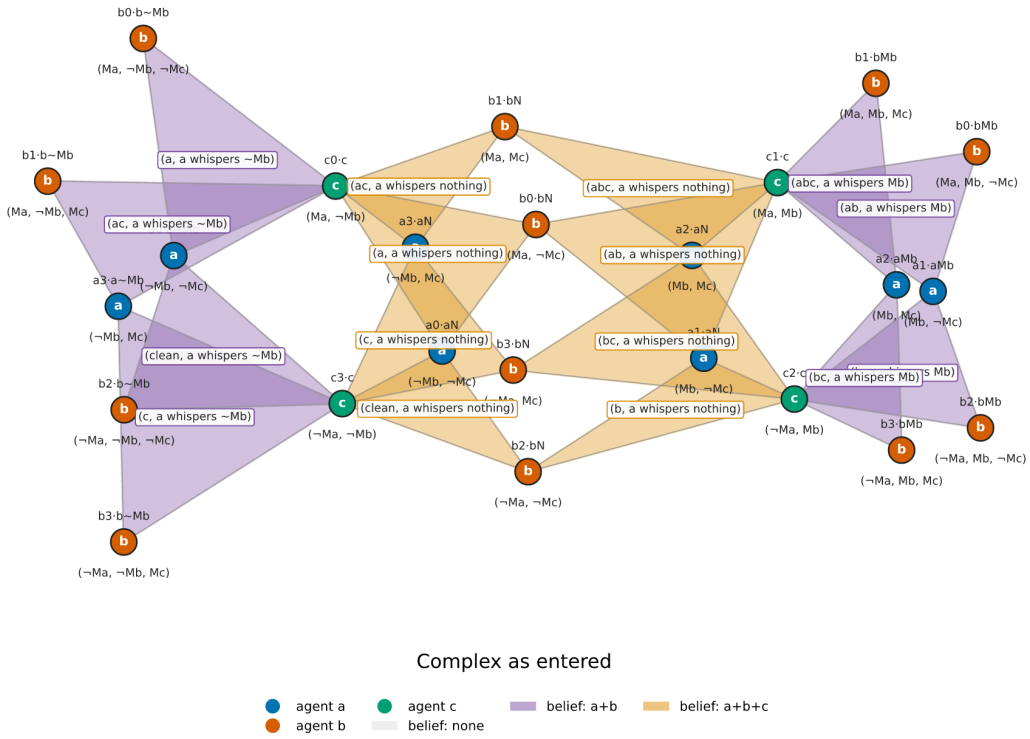
Observe: c has false beliefs in this action.

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Observe: Despite not having false beliefs in the input, c has false beliefs in the output.

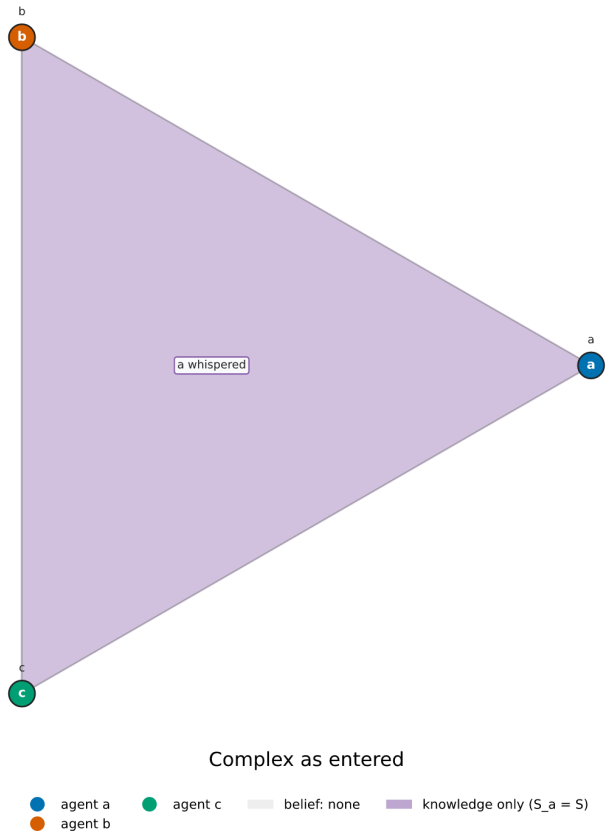
Actions with Revision

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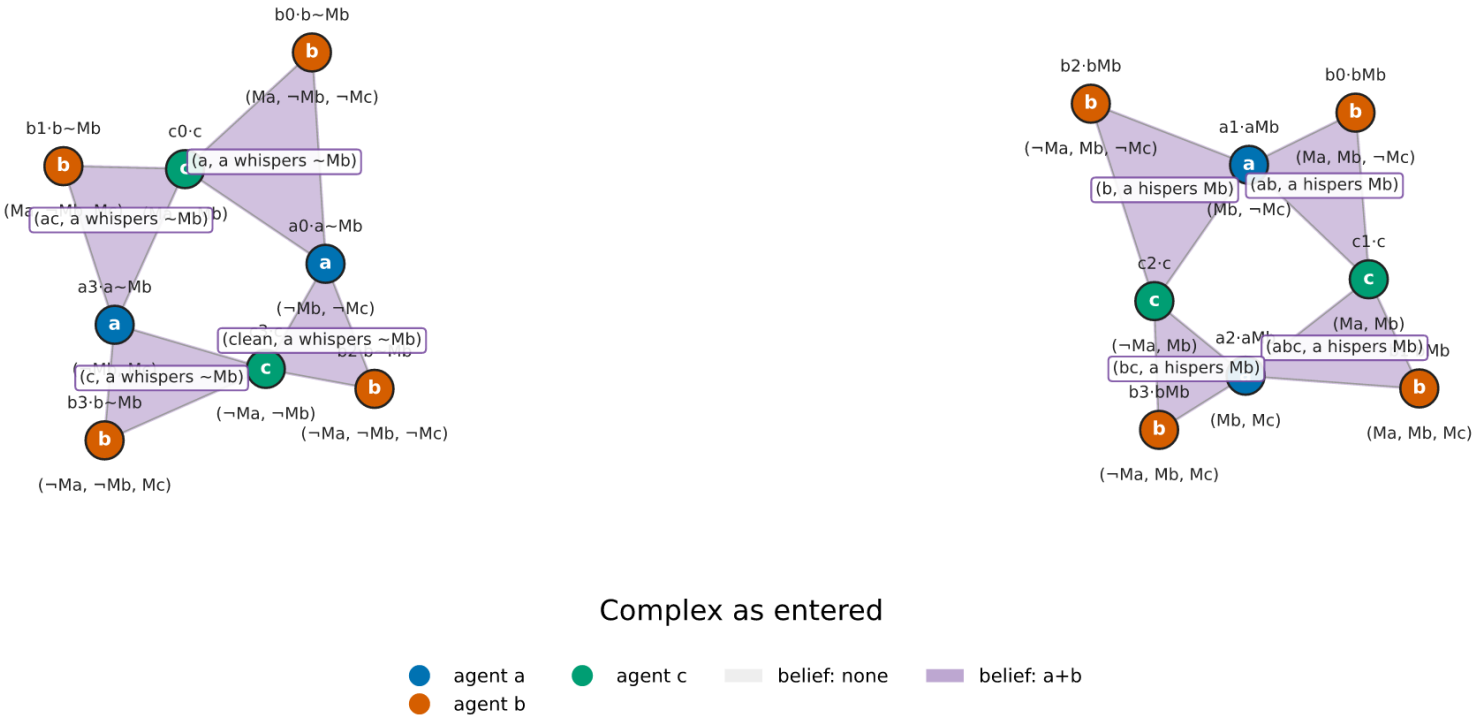
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We already know what a public announcement looks like:



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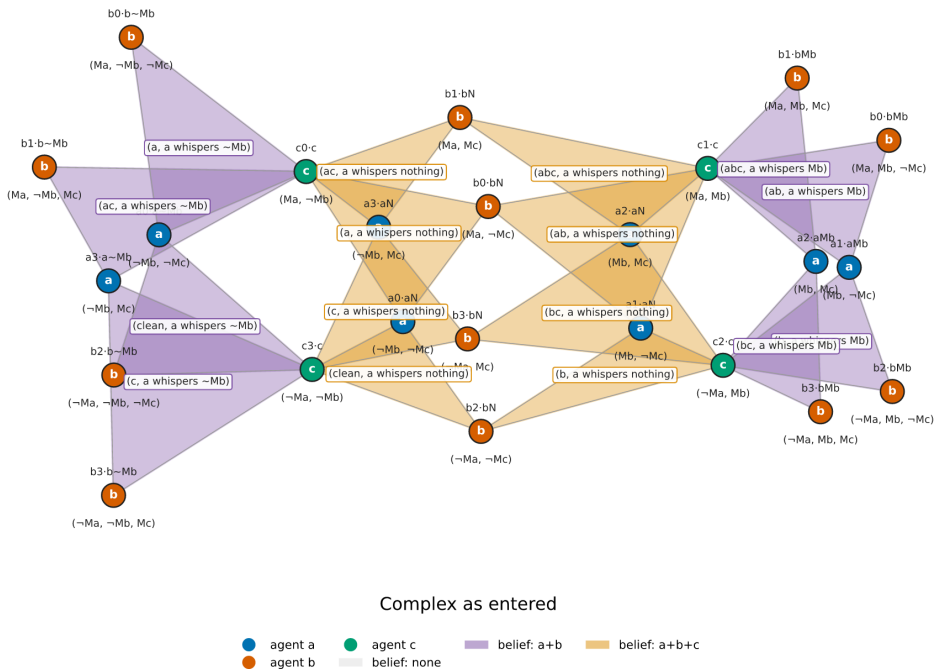
And we can compute the output:



Observe: c doesn't just have false beliefs, she believes nothing (or, equivalently, anything)!

Actions with Revision

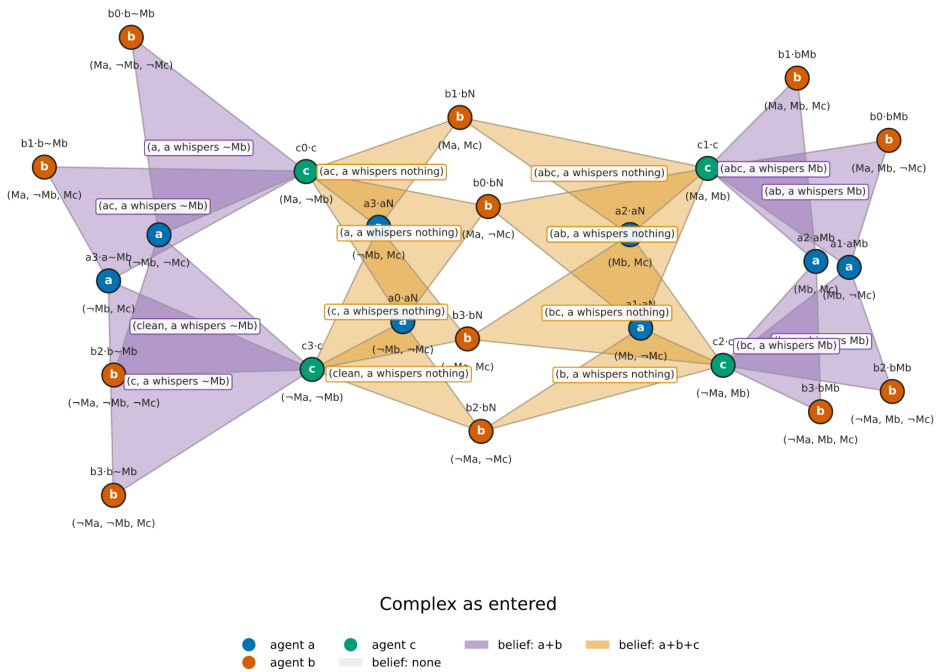
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c can "replace" lost sets with "similar" ones.[Lew76; Sta81; AGM85]

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c can "replace" lost sets with "similar" ones.[Lew76; Sta81; AGM85]

One metric for "similarity" is to say that two sets are more similar if they share more perspectives in common.

Actions with Revision

This computation is a (lot) more complicated in general. But, it gives the following output.

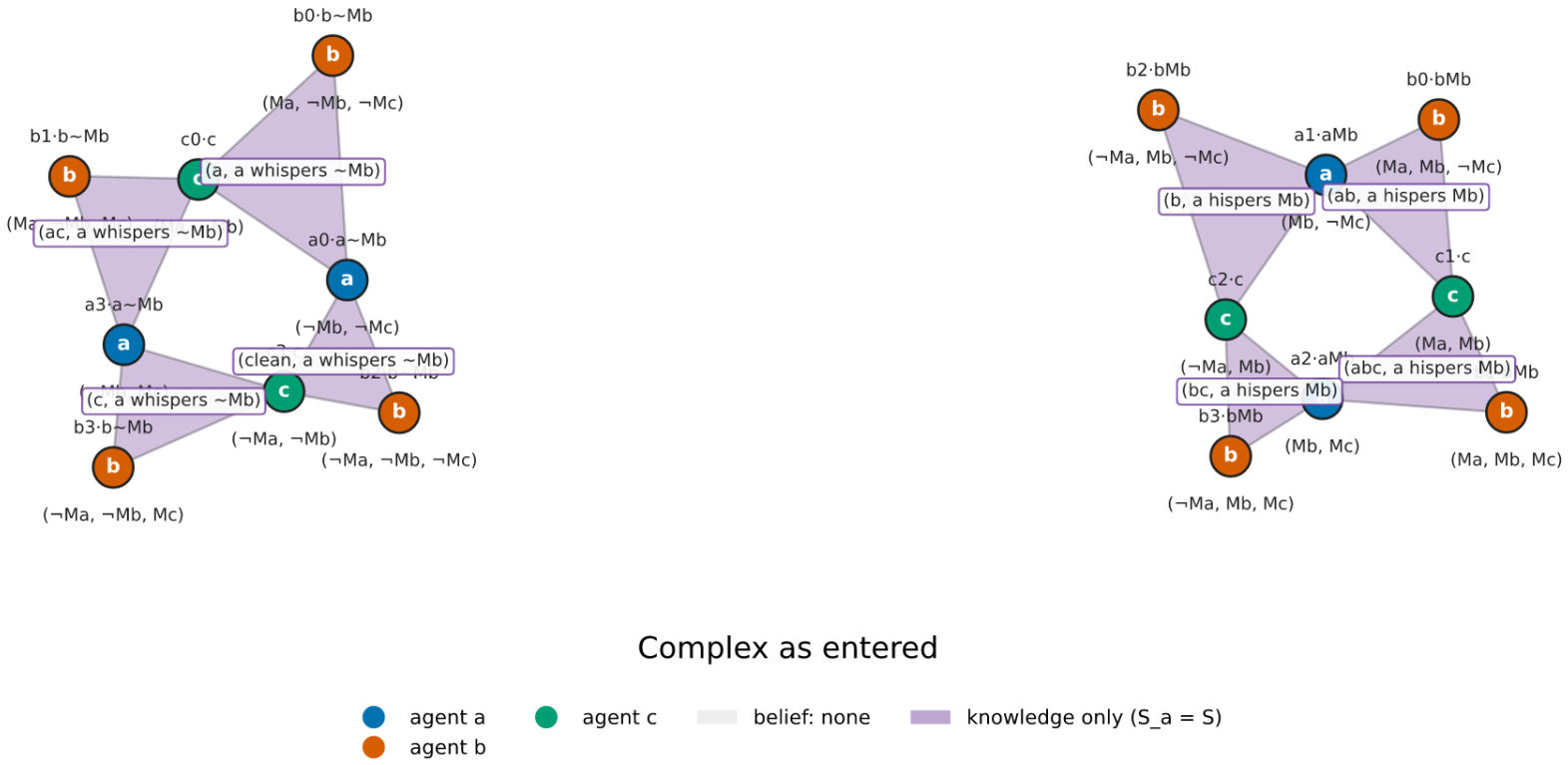


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- How does revision affect or change many existing, traditional protocols?
- Comparing the same protocol across different initial configurations, can we find protocols which don't succeed *always* but do well *typically*? Could this be a defense for some known but generally discarded protocols?

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Thank You!

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Disadvantages

- The following are sound for any $P \in \mathfrak{P}$:

$$\mathbf{NU_K} : P \to \bigvee_{a \in Ag} K_a P$$

$$\mathbf{NU_B} : P \to \bigvee_{a \in Ag} B_a P$$

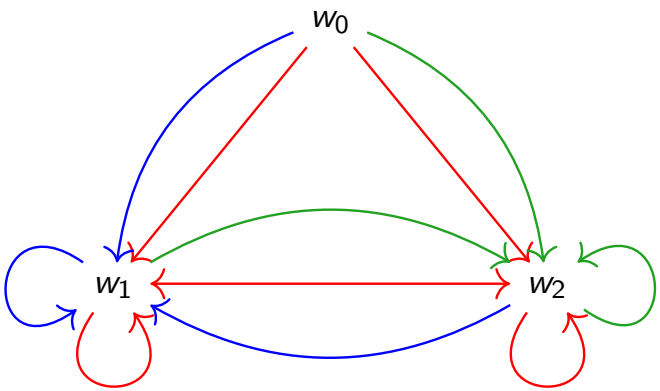
- This is because we chose to assign atoms to nodes, rather than facets
- However, **NU** is slightly more general than similar formulas in the literature [Dit+22]

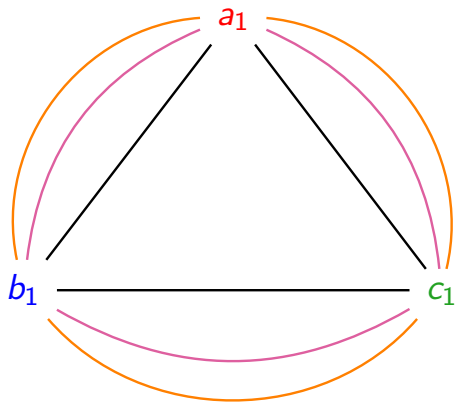
Comparison with Previous Work

- Semantics for belief in simplicial complexes has been tried twice before [Dit+22; CLS25]
- Our approach has some clear advantages, however.
- In [Dit+22], the axiom $K_a\varphi \to B_a\varphi$ is not sound
- If one restricts to models where this is sound, the stronger statement $K_a\varphi \leftrightarrow B_a\varphi$ becomes sound
- In [CLS25], the models are not UCF
- If one restricts to UCF models, $K_a\varphi \leftrightarrow B_a\varphi$ becomes sound
- Our system is the only simplicial semantics for belief we are aware of that is:
  - UCF
  - $K_a\varphi \to B_a\varphi$ is sound
  - $K_a\varphi \leftrightarrow B_a\varphi$ is not sound

Simplicial Sets

- It would be nice to avoid making axioms like **NU** sound
- There are many possible ways to do this, including "environment" nodes or hypergraphs [GKL24]
- One promising avenue is to use "simplicial sets" as opposed to simplicial complexes
- This allows multiple faces to be drawn on the same set of vertices





Awareness

- Key difference between simplicial models and Kripke models is that some formulas are "assigned" to an agent at a world/facet
- We could formalize this
- Let A_i be a new modality for $i \in Ag$ which applies to literals
- $\mathcal{M}, X \vDash A_a P$ iff $L(\pi_a(X))(P) = 1$
- Can read as "a is aware of P", similar to [**Awareness**]
- Also motivates trying to get more complicated formulas (e.g., not literals) into perspectives

Belief Merging

- The idea is still vague. But, there is an intuition that we can we can associate perspectives with belief sets
- Taking two perspectives (for different agents) could we use our notion of "nearness" to find a "best way" to combine them?
- Consider two perspectives x and y, and Let $\mathcal{X}$ and $\mathcal{Y}$ be the set of facets containing x and y respectively
- If agents beliefs come from a set of facets W, traditional merging is $\mathcal{X} \cap \mathcal{Y} \cap W$
- Perhaps $X \in \mathcal{X} \cap W$ but $X \notin \mathcal{Y}$
- Yet, there is $X' \notin W$ such that $X' \in \mathcal{X} \cap \mathcal{Y}$ and X and X' are near.
- Perhaps X' should be included in merging?

Conceptual Example

- Concern: What if we're really counting propositions, not perspectives?
- Let D be "there are ducks in the pond" and F be "there are frogs in the pond"
- For $x \in Ag \setminus \{a\}$, $x_{i,j}$ is a new node such that $L(x_{i,j})(D) = i$ and $L(x_{i,j})(F) = j$
- Initially, you consider two worlds/sets possible:

$$Y := \{a', b_{1,1}, c_{1,1}, d_{1,1}\}$$

$$Z := \{a', b_{1,0}, c_{1,2}, d_{1,2}\}$$

- You learn F
- Nearest replacement for Z is $Z' := \{a', b_{1,1}, c_{1,2}, d_{1,2}\}$, counting perspectives
- What about $Z'' := \{a', b_{1,2}, c_{1,1}, d_{1,2}\}$?
- Z' and Z'' satisfy the same formulas, but Z' only differs from Y by a single perspective, as opposed to two

Shifting from Knowledge to Belief

- You might try designating a UCF subcomplex of S, call it S', as the worlds the agents consider possible
- Fix a model $\mathcal{M}$ and $X \in \mathcal{F}(S')$
- $\mathcal{M}, X \models B_a\phi$ iff $\forall Y \in \mathcal{F}(S')$ if $\pi_a(Y) = \pi_a(X)$ then $\mathcal{M}, Y \models \phi$
- This makes the following sound for all $a, b \in Ag$: $\mathcal{M} \models B_a(B_b\phi \to \phi)$

Belief ala Chapter 2

- Let $\mathfrak{P}$ be a countable set of propositional atoms
- A **simplicial knowledge and belief model** $\mathcal{M}$ is a simplicial frame $(N, V, , S, \{S_a\}_{a \in Ag}, L)$ together with a *valuation function* $L : \mathfrak{P} \to 2^{\mathcal{F}(S)}$
- Intuitively, $L(P)$ tells us which facets P is true at
- the language $\mathcal{L}_{KB}(Ag)$ recursively defined by

$$\varphi ::= P \mid \bot \mid \phi \to \psi \mid K_a\phi \mid B_a\phi$$

$\mathcal{M}, X \models P$ iff $X \in L(P)$

$\mathcal{M}, X \not\models \bot$

$\mathcal{M}, X \models \phi \to \psi$ iff $\mathcal{M}, X \models \phi$ implies $\mathcal{M}, X \models \psi$

$\mathcal{M}, X \models K_a\phi$ iff $\forall Y \in \mathcal{F}(S)$ if $\pi_a(Y) = \pi_a(X)$ then $\mathcal{M}, Y \models \phi$

$\mathcal{M}, X \models B_a\phi$ iff $\forall Y \in \mathcal{F}(S_a)$ if $\pi_a(Y) = \pi_a(X)$ then $\mathcal{M}, Y \models \phi$

Soundness and Completeness Theorems From Chapter 2

1. knowledge implies belief: $K_a\phi \to B_a\phi$
2. strong positive introspection: $B_a\phi \to K_aB_a\phi$
3. strong negative introspection: $\neg B_a\phi \to K_a\neg B_a\phi$

Let FULL be the axiom system obtained by adding these three axiom schemes to the system S5 for knowledge combined with KD45 for belief.

Theorem

FULL is sound and complete with respect to the class of all simplicial belief models.

Soundness and Completeness Theorems From Chapter 3

Theorem (Soundness and Completeness)

The axiom schema **K45+NU**, *plus the axioms of propositional logic and the inference rules necessitation and modus ponens are sound and complete with respect to simplicial belief models.*

Full Formal Definition of Revision

Definition

Consider a UCF simplicial model $\mathcal{M} = \langle N, V, L, \{S_a\}_{a \in A} \rangle$, and φ a formula with no modalities in its recursive construction. Let $N_\varphi := N$, $V_\varphi := V$, and $L_\varphi := L$. Further, let $\mathbf{F}_\varphi$ be the subset of $\mathcal{F}(\mathfrak{M}((N_\varphi, V_\varphi, L_\varphi))$ which satisfy φ. We need to define a function $\mathcal{R}_a : \mathcal{F}(S_a) \times \text{Ł}_B(Ag) \to 2^{\mathbf{F}_\varphi}$.

$$\mathcal{R}_a(F, \varphi) := \{G \in \mathbf{F}_\varphi \mid (\pi_a(F) = \pi_a(G)) \wedge$$

$$\forall X \in \mathbf{F}_\varphi((\pi_a(X) = \pi_a(G)) \to (|F \cap G| \geq |X \cap G|))\}$$

Note that $\mathcal{R}_a : \mathcal{F}(S_a) \times \text{Ł}_B(Ag) \to 2^{\mathbf{F}_\varphi}$ is a nearness function.[a] Further, note that, if $G \in \mathbf{F}_\varphi$, then $\mathcal{R}_a(G, \varphi) = \{G\}$. Let each $S_{a,\varphi}$ be the simplicial complex whose facets are elements of the set $\bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$. Then $\mathcal{M}[\varphi] := \langle N_\varphi, V_\varphi, L_\varphi, \{S_{a,\varphi}\}_{a \in A} \rangle$.
If S is not maximal, simply take each $S_{a,\varphi}$ be the simplicial complex whose facets are elements of the set $\mathcal{F}(S) \cap \bigcup_{F \in \mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$.

Things Go Wrong Without Memory

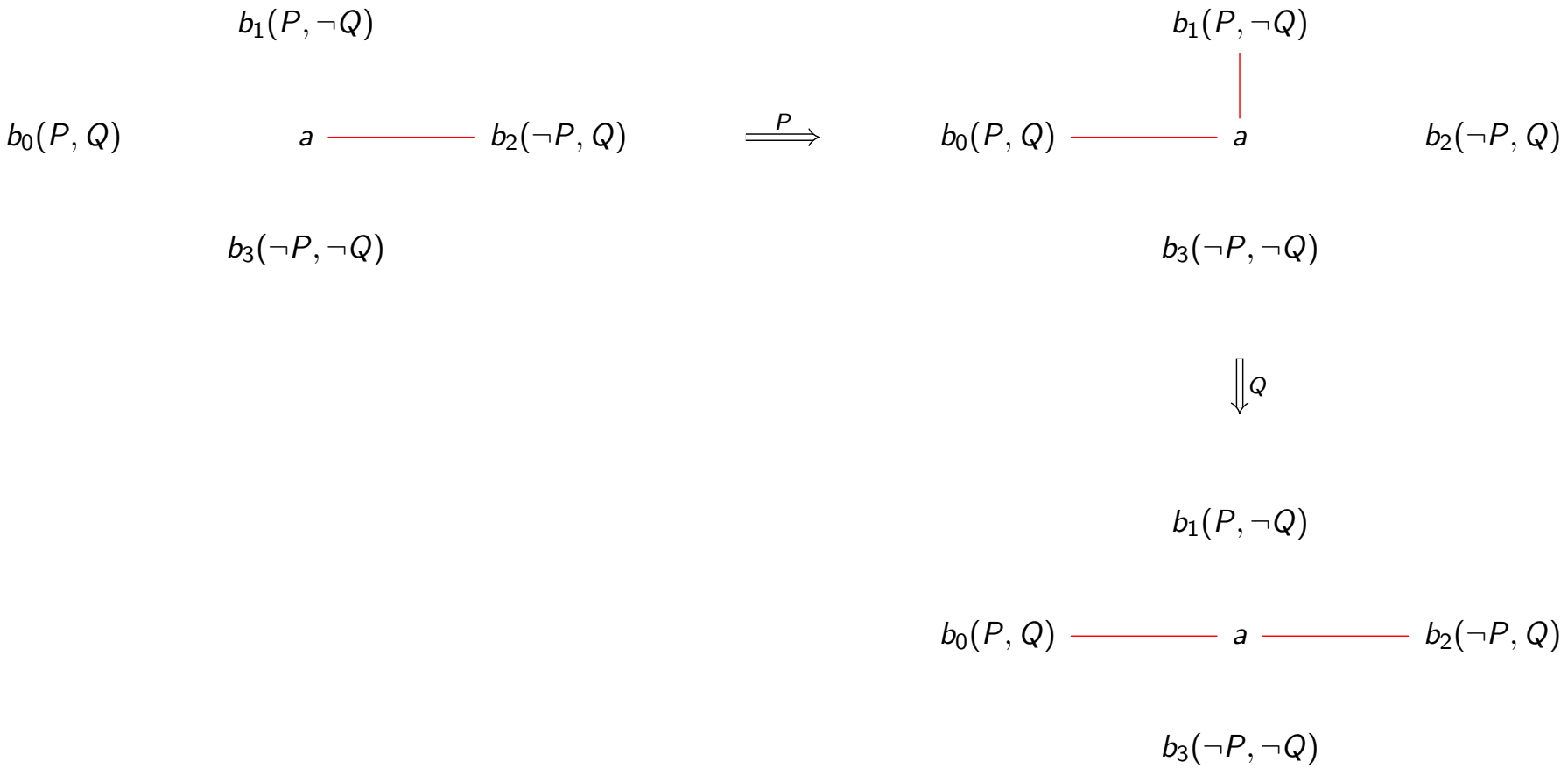


Figure: Three models with two agents, showing an initial model with four facets in $W = \mathcal{M}(N, V, L)$, corresponding to all possible arrangements of the atoms P and Q. There are two sequential updates. First the agent updates by P and then Q. Agent A begins by believing $\neg P \wedge Q$.

Revision With Memory

Definition

Consider a UCF simplicial model $\mathcal{M} = \langle N, V, L, S, \{S_a\}_{a\in A}\rangle$, and φ a formula with no modalities in its recursive construction.. Let $N_\varphi := N$, $V_\varphi := V$, $L_\varphi := L$. Further, let $\mathbf{F}_\varphi$ be the subset of $\mathcal{F}(\mathfrak{M}((N_\varphi, V_\varphi, L_\varphi))$ which satisfy φ. Also let S_φ be the simplicial complex whose facets are given by $\mathcal{F}(S) \cap \mathbf{F}_\varphi$. We need to define a function $\mathcal{R}_a : \mathcal{F}(S_a) \times Ł_B(Ag) \to 2^{\mathbf{F}_\varphi}$.

$$\mathcal{R}_a(F, \varphi) := \{G \in \mathcal{F}(S_\varphi) \mid (\pi_a(F) = \pi_a(G))$$

$$\wedge \forall X \in \mathbf{F}_\varphi((\pi_a(X) = \pi_a(G)) \to (|F \cap G| \geq |X \cap G|))\}$$

Note that $\mathcal{R}_a : \mathcal{F}(S_a) \times Ł_B(Ag) \to 2^{\mathbf{F}_\varphi}$ is a nearness function. Further, note that, if $G \in \mathcal{F}(S_\varphi)$, then $\mathcal{R}_a(G, \varphi) = \{G\}$. Let each $S_{a,\varphi}$ be the simplicial complex whose facets are elements of the set $\bigcup_{F\in\mathcal{F}(S_a)} \mathcal{R}_a(F, \varphi)$. Then $\mathcal{M}[\varphi]_m := \langle N_\varphi, V_\varphi, L_\varphi, S_\varphi, \{S_{a,\varphi}\}_{a\in A}\rangle$.

Things Don't Go Wrong With Memory

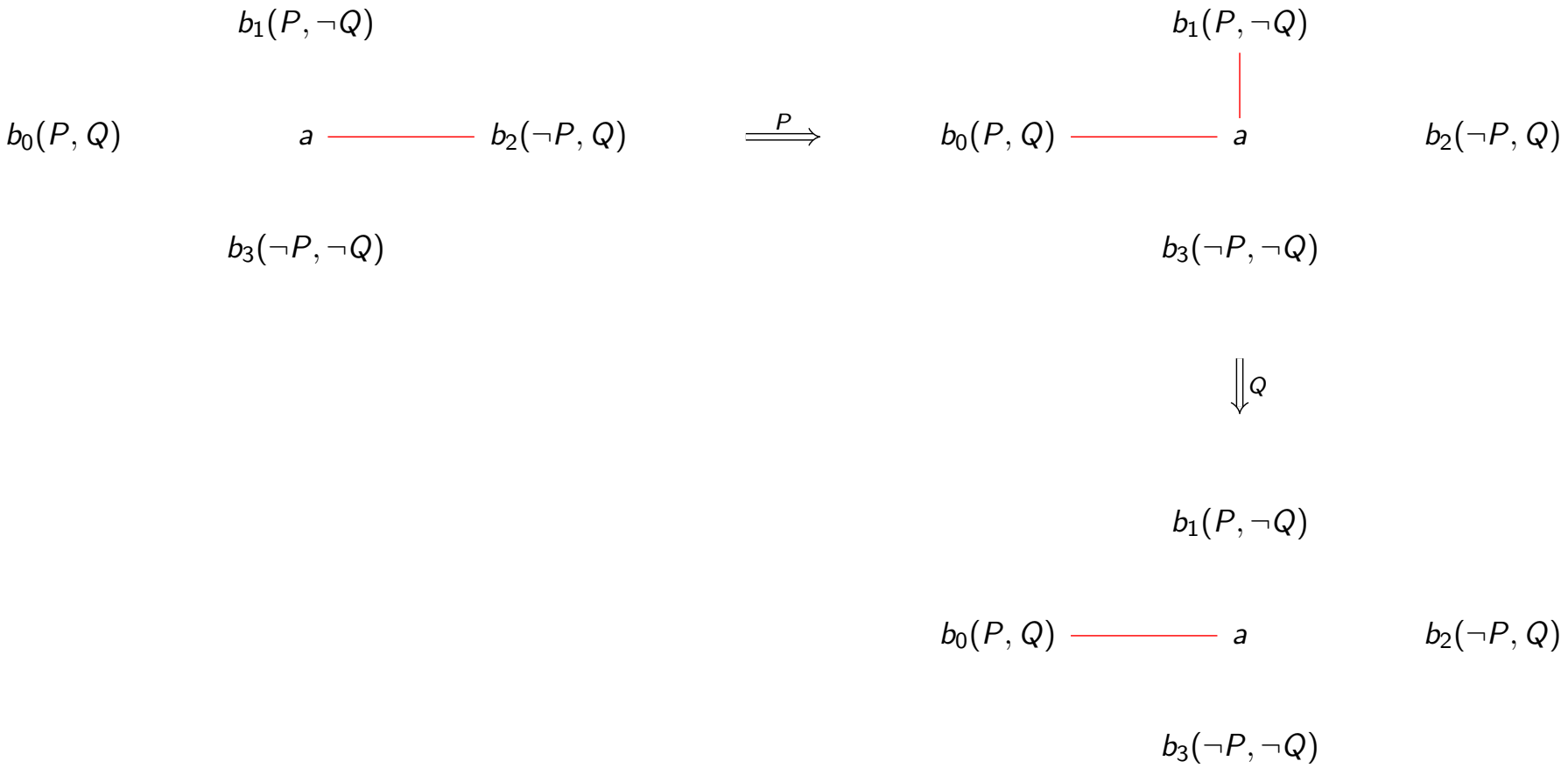


Figure: S_c before and after P_a is publicly announced, assuming revision happens only when a perspective is no longer contained in any world.

Node Modes

Our definition of $N[A]$ is not the only one which could be motivated. In fact, all three of the definitions below are defensible:

1. $\forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) = 1) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) = 0))$
2. $\forall P \in \mathfrak{P}((L_A(y)(P) = 1 \rightarrow L(x)(P) = 1) \wedge (L_A(y)(P) = 0 \rightarrow L_A(x)(P) = 0))$
3. $\forall P \in \mathfrak{P}((L(x)(P) = 1 \rightarrow L_A(y)(P) \neq 0) \wedge (L(x)(P) = 0 \rightarrow L_A(y)(P) \neq 1))$

Mode 3 is sufficient, but this is worth exploring further.

Knowledge Action

Definition (Simplicial Action)

Given a simplicial model for knowledge $\mathcal{M} = \langle N, V, L, S\rangle$, And a simplicial action $A = \langle N_A, V_A, L_A, S_A, Post\rangle$, we define the update model $\mathcal{M}[A] := \langle N[A], V[A], L[A], S[A]\rangle$ as follows:

$$\begin{aligned} N[A] &:= \{(x,y) \in N \times N_A | V(x) = V_A(y)\} \\ &\quad \wedge \forall P \in \mathfrak{P}((L(x)(P) = 1 \to L_A(y)(P) \neq 0) \\ &\quad \wedge (L(x)(P) = 0 \to L_A(y)(P) \neq 1))\} \\ V[A]((x,y)) &:= V(x) \end{aligned}$$

$$L[A]((x,y))(P) := \begin{cases} Post(y)(P) \text{ if } Post(y)(P) \neq 2 \\ L_A(y)(P) \text{ if } L(x)(P) = 2 \\ L(x) \text{ otherwise} \end{cases}$$

$$\mathcal{F}(S[A]) := \left\{ F \in 2^{N[A]} \;\middle|\; \begin{array}{c} p_0(F) \in \mathcal{F}(S) \wedge p_1(F) \in \mathcal{F}(S_A) \\ \wedge F \text{ is consistent w.r.t. } L[A] \end{array} \right\}$$

Belief Action

Definition (Simplicial Action for Belief)

Given a simplicial model for belief $\mathcal{M} = \langle N, V, L, S\rangle$, And a simplicial action for belief $A = \langle N_A, V_A, L_A, \{S_{a,A}\}_{a\in Ag}, Post\rangle$, we define the update model $\mathcal{M}[A]_B := \langle N[A]_B, V[A]_B, L[A]_B, \{S_a[A]_B\}_{a\in Ag}\rangle$ as follows:

$$\begin{aligned} N[A]_B &:= \{(x,y) \in N \times N_A | V(x) = V_A(y)\} \\ &\quad \wedge \forall P \in \mathfrak{P}((L(x)(P) = 1 \to L_A(y)(P) \neq 0) \\ &\quad \wedge (L(x)(P) = 0 \to L_A(y)(P) \neq 1))\} \\ V[A]_B((x,y)) &:= V(x) \end{aligned}$$

$$L[A]_B((x,y))(P) := \begin{cases} Post(y)(P) \text{ if } Post(y)(P) \neq 2 \\ L_A(y)(P) \text{ if } L(x)(P) = 2 \\ L(x) \text{ otherwise} \end{cases}$$

$$\mathcal{F}(S_a[A]_B) := \left\{ F \in 2^{N[A]_B} \;\middle|\; \begin{array}{c} p_0(F) \in \mathcal{F}(S_a) \wedge p_1(F) \in \mathcal{F}(S_{a,A}) \\ \wedge F \text{ is consistent w.r.t. } L[A]_B \end{array} \right\}$$

Belief Action with Revision

Definition

$N[A]_{BR} := N[A]_B$, $V[A]_{BR} := V[A]_B$, and $L[A]_{BR} := L[A]_B$. Defining $S_a[A]_{BR}$ will take some more work. Fix $a \in Ag$ and let X be a UCF facet in $2^{N \times N_A}$ such that $p_0(X) \in \mathcal{F}(S_a)$ and $p_1(X) \in \mathcal{F}(S_{a,A})$. Call the set of such X $\mathcal{F}^a(2^{N[A]_{BR}})$. Let X be a UCF facet in $2^{N[A]_{BR}}$ such that X is consistent, and $p_1(X) \in \mathcal{F}(S_{a,A})$. Call the set of such X $\mathcal{F}_a(2^{N[A]_{BR}})$. Then we can define a replacement function as follows for $X \in \mathcal{F}^a(2^{N \times N_A})$:

$$R_a(X) := \{Y \in \mathcal{F}_a(2^{N[A]_{BR}}) | \pi_a(Y) = \pi_a(X)$$

$$\wedge(\forall Z \in \mathcal{F}_a(2^{N[A]_{BR}})(\pi_a(Z) = \pi_a(X) \rightarrow |Y \cap X| \geq |Z \cap X|))\}$$

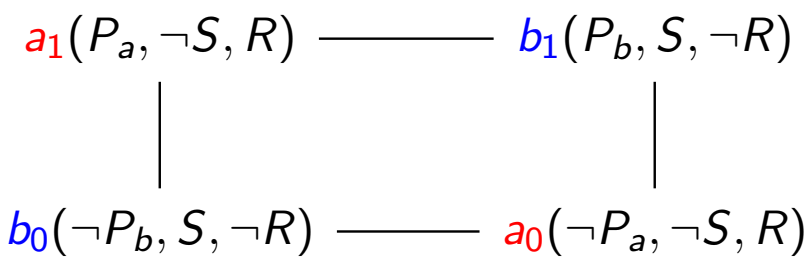
We say that $S_a[A]_{BR}$ is defined as the simplicial complex whose facets come from $\bigcup_{X \in \mathcal{F}^a(2^{N \times N_A})} R_a(X)$.

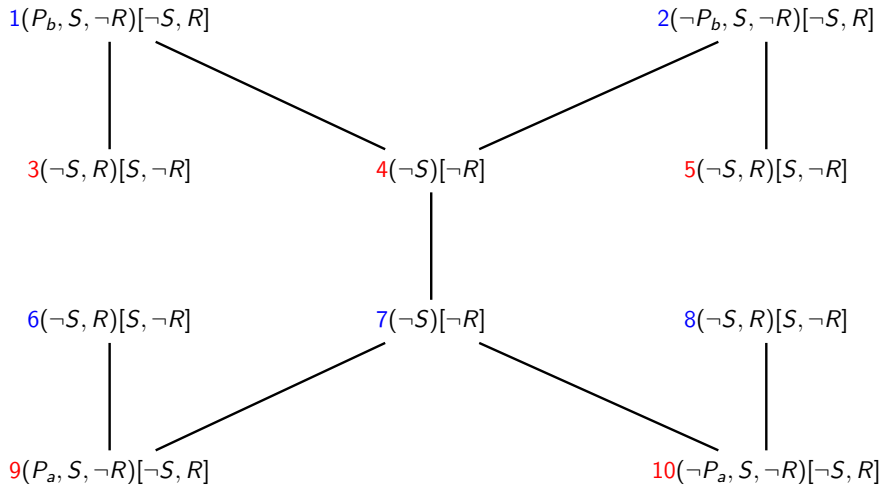
Inverse Revision Function

For any face X in $S_a[A]_{BR}$:

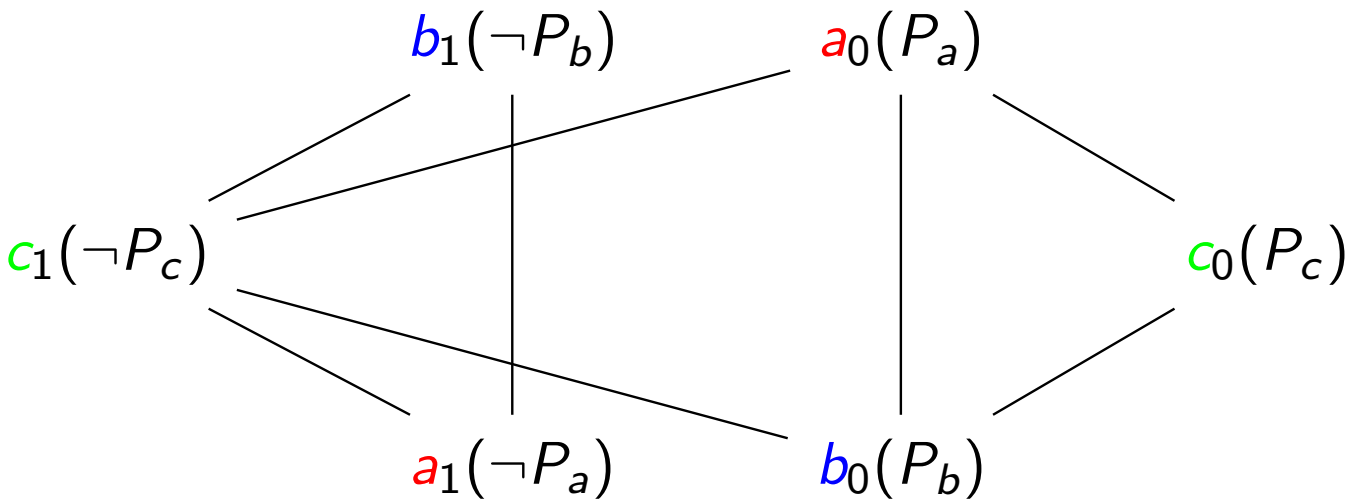
$$R_a^{-1}(X) := \{X' \mid \exists Y \in \mathcal{F}(S_a[A]_{BR})(X \subseteq Y \wedge \exists Y' \in R_a^{-1}(Y)(X' \subseteq Y'))$$
$$\wedge \forall a \in Ag(\exists x \in X(V[A]_{BR}(x) = a)$$
$$\leftrightarrow \exists x' \in X'(V[A]_{BR}(x') = a))\}$$

Alternating Message Passing Protocol

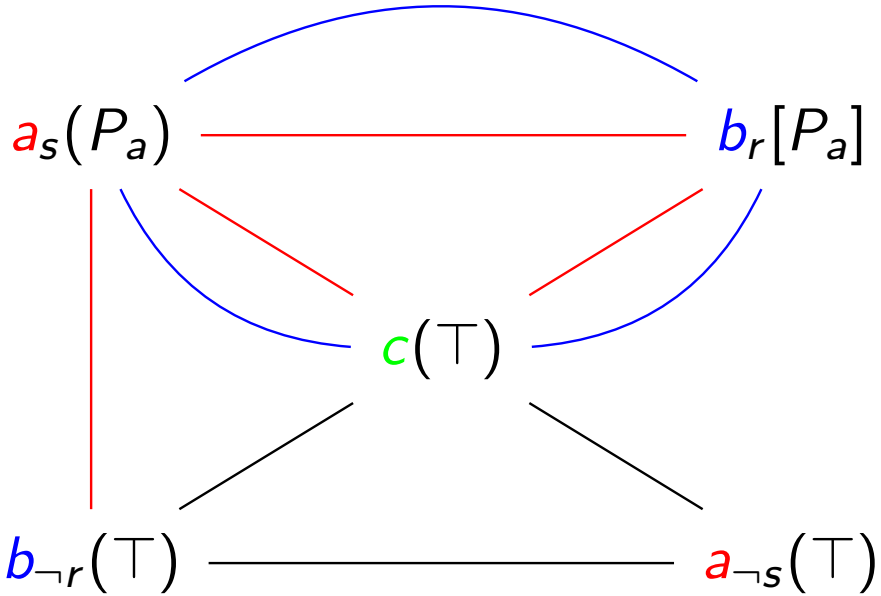




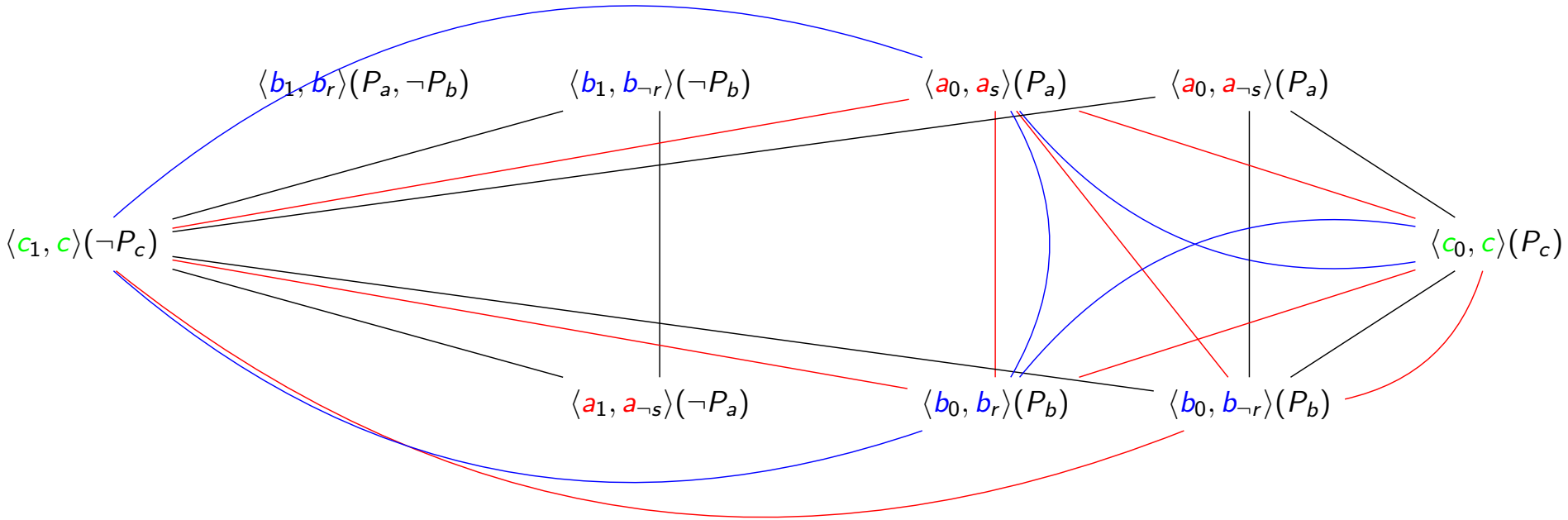
Example of Belief Action (Input)



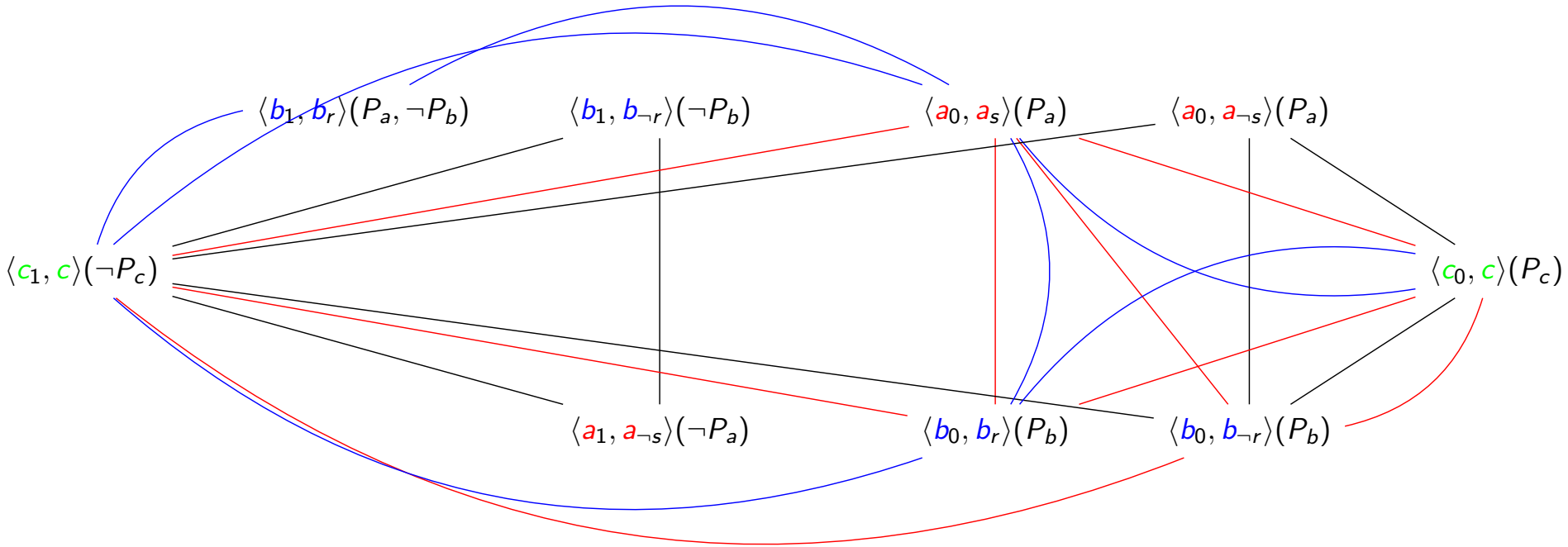
Example of Belief Action (Action)



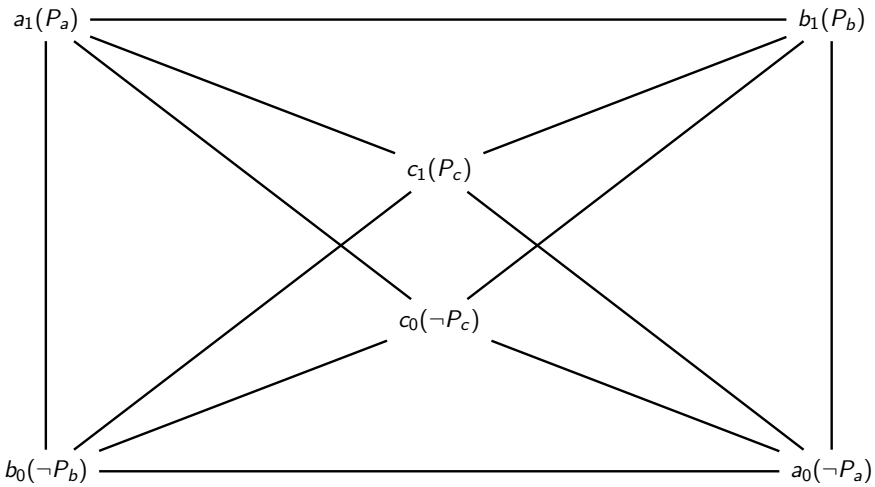
Example of Belief Action (No Revision) (Output)

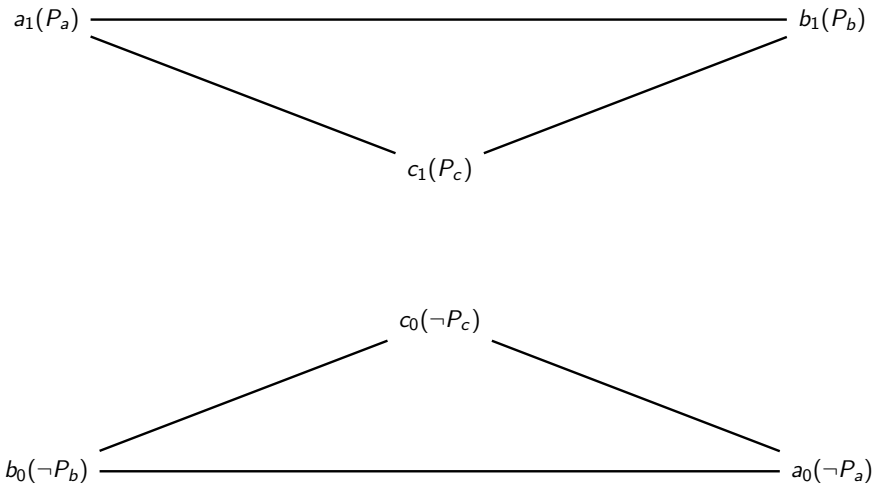


Example of Belief Action with Revision (Output)

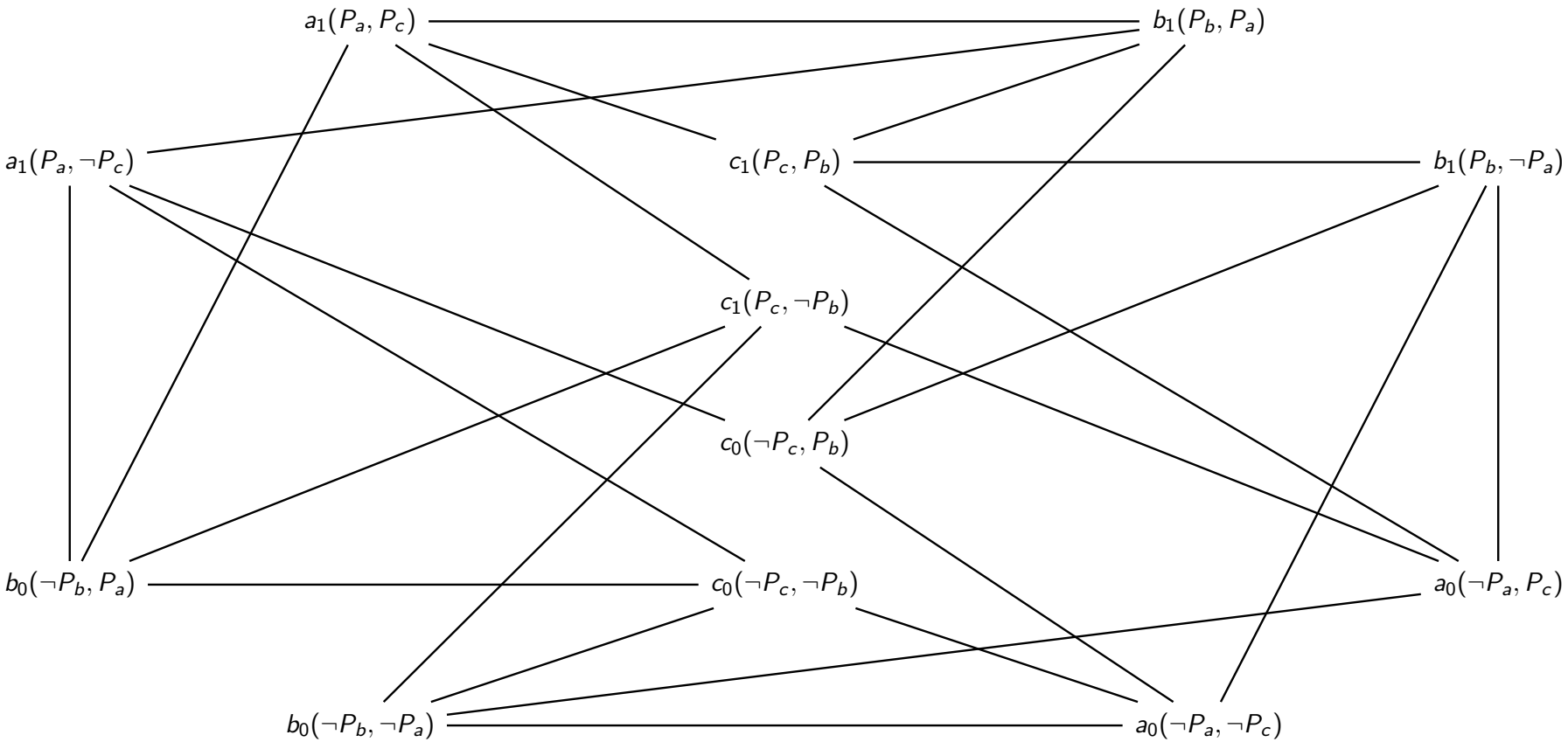


Example of Leveraging Topology Task

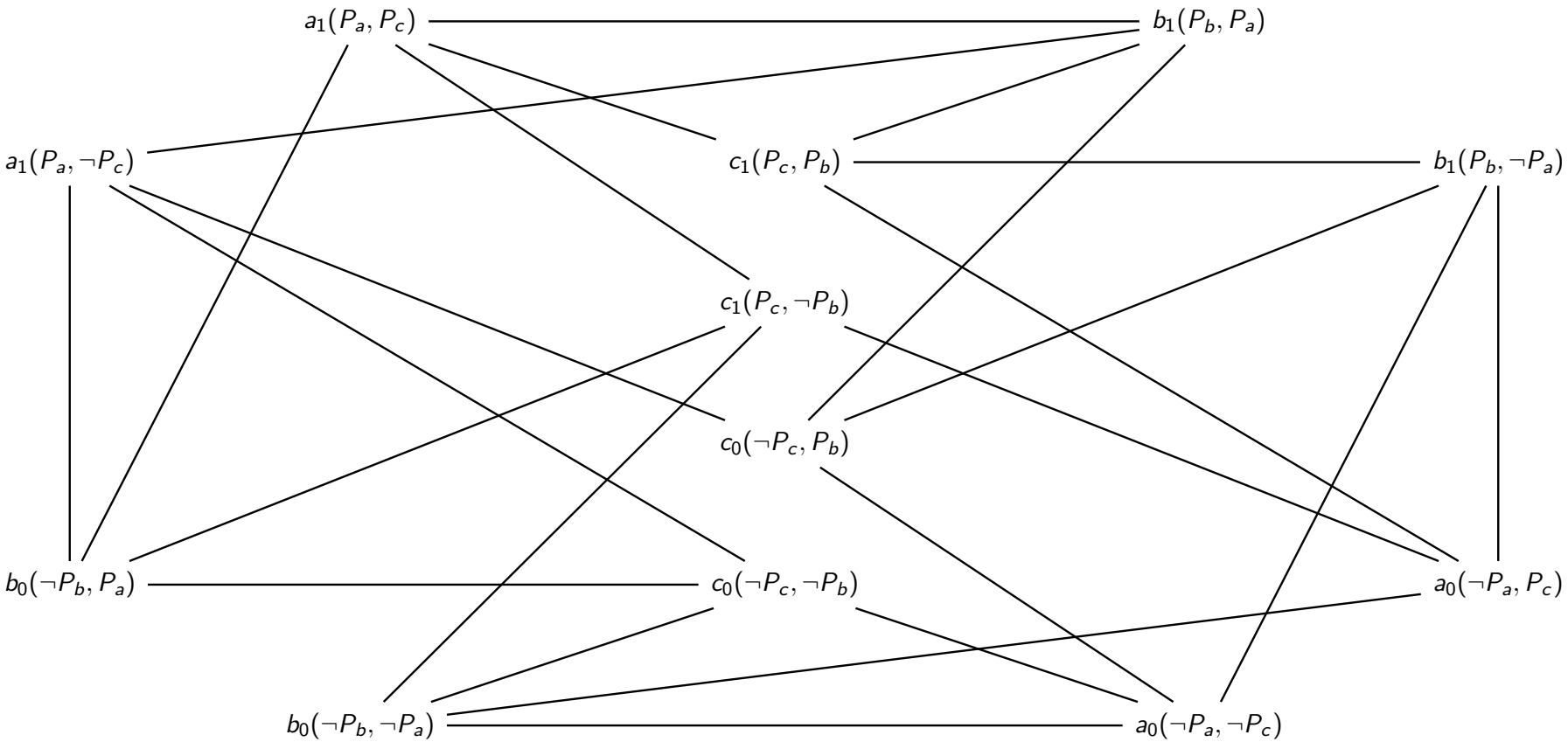




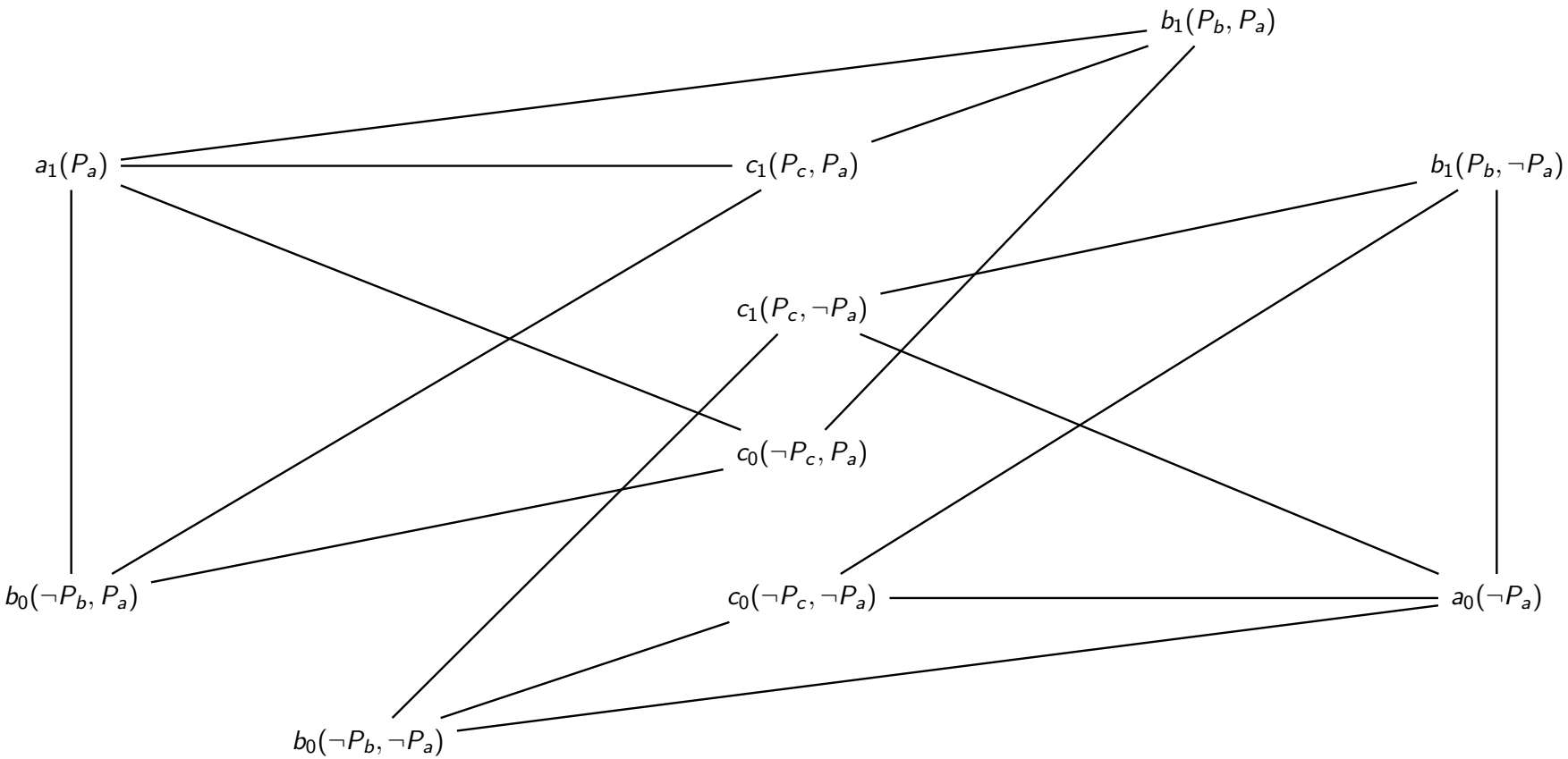
Example of Leveraging Topology
Signal Sequence 1: Previous Approach (Application)



Example of Leveraging Topology
Signal Sequence 1: Action



Example of Leveraging Topology
Signal Sequence 2: Previous Approach (Application)



Example of Leveraging Topology
Signal Sequence 2: Action

